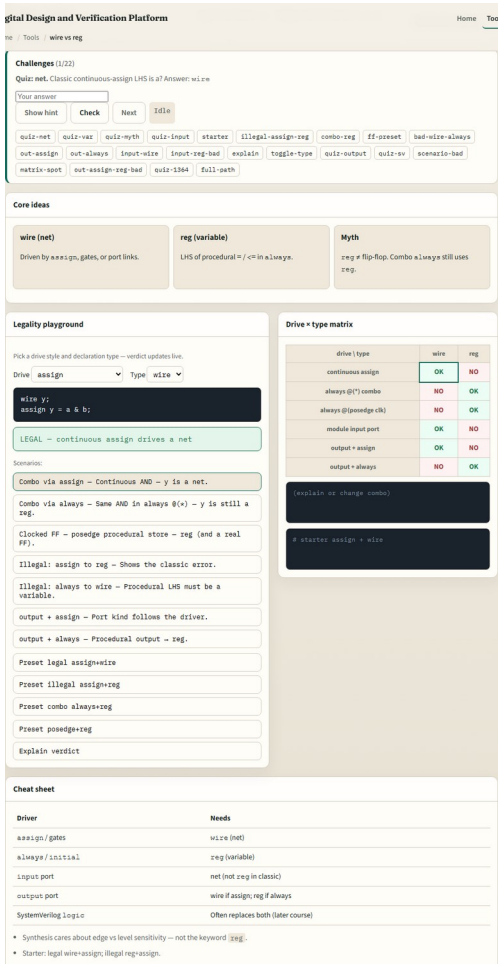
wire vs reg

Classic Verilog splits names into two families: nets, declared with wire, and variables, declared with reg

Nets versus variables

- A wire is a net
- You drive it with continuous assign
- A reg is a variable
- You drive it inside procedural blocks
- Input ports are nets in IEEE 1364
- Output can be a net if assign drives it, or a reg if always drives it

Browser lab



Wire vs reg lab starter

Real Verilog practice

wsl – module03-wire-vs-reg/examples/navigation

```
# Track A – real Unix
# real Verilog session (Track A)

$ cat examples/wire_reg/wire_vs_reg.v
// wire_vs_reg.v – assign-driven net vs always-driven variable
module wire_reg_demo(
  input  a,
  input  b,
  output y_net,
  output reg y_var
);
  assign y_net = a & b;
  always @(*) y_var = a & b;
endmodule

$ iverilog -t null examples/wire_reg/wire_vs_reg.v
(syntax check passed)
```

Real Verilog — wire and reg drivers

Pitfalls to watch

- Do not read reg as “register hardware”, combo always blocks still need a reg on the left
- Do not continuous-assign a reg or procedural-assign a wire
- SystemVerilog logic blurs the picture in later courses
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and finish a few challenges
- On real Verilog
- When you are ready